VINITH KUMAR T

AUTONOMOUS ROBOTICS ENGINEER

Dharmapuri, Tamil nadu | vinithkumart2002@gmail.com | **PORTFOLIO -** https://vinithkumar-themorgan.github.io/

SUMMARY

Autonomous Robotics Engineer specializing in end-to-end robotic system development, from mechanical design and fabrication to electronics, embedded systems, ROS2, autonomy, integration, and testing. Experienced in building autonomous robots from scratch and developing navigation, perception, and control systems. Seeking a multidisciplinary robotics role focused on designing, integrating, and deploying complete autonomous systems.

KEY ACHIEVEMENTS

- **KAVACH 2023** — National Winner (Govt. of India)
- **SIH 2022** — 2nd Runner-Up (Govt. of India)
- **Patent Applied** — Six-Axis Pick-and-Place Robot
- **Publication** — Quadruped Robot, IRE Journals
- **Flipkart GRID 5.0** — National Finalist (Robotics)
- **Flipkart GRID 4.0** — Excellence Award (Drones)
- **Hackfest 2023** — Finalist (Got internship offer)
- **Aerathon 2023 (SAE)** — 23rd, National

PROFESSIONAL EXPERIENCE

Robotics Engineer – Full-Stack Robotics & Autonomous Systems | Robolog Automation
June 2024 - Present

- Built 3 autonomous robots from scratch for EdTech and networking applications, spanning mechanical design, fabrication, electronics, embedded systems, ROS2, and testing.
- Secured a ₹1-crore Tamil Nadu government project by successfully showcasing two EdTech AMR platforms (ROS & ROS2) during a competitive project allocation round against multiple companies.
- Designed a dynamic multi-controller architecture that switches controllers based on navigation objectives, achieving ~5 mm positional accuracy during path-following and goal-reaching tests.
- Designed and wired industrial control panels and developed automated hardware test jigs for validating BMS circuits, motor drivers, and other electronic modules.
- Led the end-to-end development lifecycle, from concept and CAD to fabrication, wiring, software integration, and final testing.

PROJECTS

Bannari Amman Institute of Technology
Sep 2020 - May 2024

Autonomous Security Mobile Robot | ROS · SLAM · Sensor Fusion · AI Perception |

- Developed an autonomous mobile robot integrating ROS navigation, SLAM, sensor fusion, and AI-based perception for autonomous security applications.

Autonomous Package Delivery Drone | Pixhawk · Computer Vision · ArUco · Depth Camera |

- Developed an autonomous package-delivery drone using Pixhawk, depth vision, and ArUco-based landing/target detection.

6-Axis Vision-Guided Robotic Manipulator | Robotic Manipulation · Computer Vision · Automation |

- Developed a vision-guided robotic manipulation system for autonomous package identification and sorting.

Autonomous Mobile Manipulator | ROS · Raspberry Pi · Vision · Robotics |

- Developed a mobile robotic platform integrating robotic manipulation and vision-based navigation.

mmWave Vital-Sign Sensing / Waveguide | RF · Waveguide Design · mmWave |

- Designed and analyzed a waveguide-based mmWave sensing system for non-contact vital-sign monitoring.

Additional Projects: 7+ projects spanning autonomous drones, robotics, computer vision, embedded systems, industrial automation, RF/mmWave, and mechanical design. **View more projects → LinkedIn | Portfolio**

ROBOTICS LEADERSHIP & R&D EXPERIENCE

Team Lead & Full-Stack Robotics Developer — Team Little Champs

Bannari Amman Institute of Technology | Sep 2020 – May 2024

- Led a multidisciplinary robotics team from concept and development through system integration, testing, and national-level competitions, developing AMRs, UAVs, robotic manipulators, and AI/embedded systems while mentoring students in robotics and autonomous system development.

Robotics Engineer – R&D (Volunteer)

Bannari Amman Institute of Technology | Mar 2024 – May 2024

- Programmed and integrated a ROS-based AMR for autonomous navigation and demonstrated the system at INTEC Expo 2024.

TECHNICAL SKILLS

Robotics & Autonomy:

- ROS / ROS2, SLAM, Nav2, Navigation, Sensor Fusion, Perception, Forward & Inverse Kinematics, Foxglove Studio

Computer Vision & AI:

- OpenCV, YOLO, Roboflow, Deep Learning, Machine Learning, Object Detection

Programming:

- Python, C++

Embedded & Electronics:

- ESP32, STM32, UART, I^2C, SPI, CAN, RS-485, EtherCAT, Sensor/Actuator Integration, Encoder Integration, Hardware Testing

Drones & RF:

- Pixhawk, PX4, ArduPilot, Drone Design, Waveguide Design, RF Analysis, mmWave

Mechanical & Fabrication:

- SolidWorks, Onshape, CAD, Mechanical Design, Gears & Transmissions, Sheet Metal, 3D Printing, MIG Welding, Laser/CNC, Lathe & Milling , FEA, CFD

Industrial Automation & IoT:

- MQTT, Node-RED, HMI Design, Control Panel Design & Wiring

EDUCATION

Bachelor of Electronics and Instrumentation Engineering — **Sep 2020 - May 2024**

- Bannari Amman Institute of Technology — CGPA : 7.9

ADDITIONAL INFORMATION

Languages: English, Tamil.